Learning Theory

Infinite Hypothesis Sets

Recap

- Last Time on Learning Theory:
  - Theoretical approach to arguing that $E_{in}(h) \approx E_{out}(h)$.



Out-of-Sample Error

- Data Generation:
  - Input domain X, output domain Y
  - Unknown distribution D over X
  - Unknown target $f : X \rightarrow Y$

- Binary 0-1 Loss:

$$E_{out}(h) = Pr_{x \sim D}[h(x) \neq f(x)]$$ f deterministic

In-Sample Error

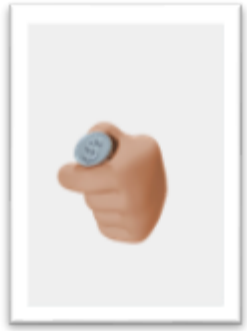
- Data Generation:
  - Input domain X, output domain Y
  - Unknown distribution D over X
  - Unknown target f : X → Y
  - Receive n samples $(x_1,f(x_1)),(x_2,f(x_2)),\ldots,(x_n,f(x_n))$.

- Binary 0-1 Loss:

$$E_{in}(h) = \frac{1}{n}\sum_{i=1}^{n} 1_{h(x_i)\neq f(x_i)}$$

f deterministic

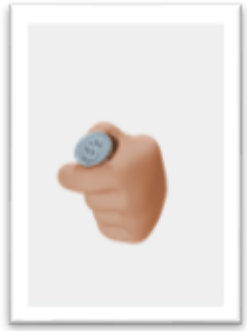
Recap



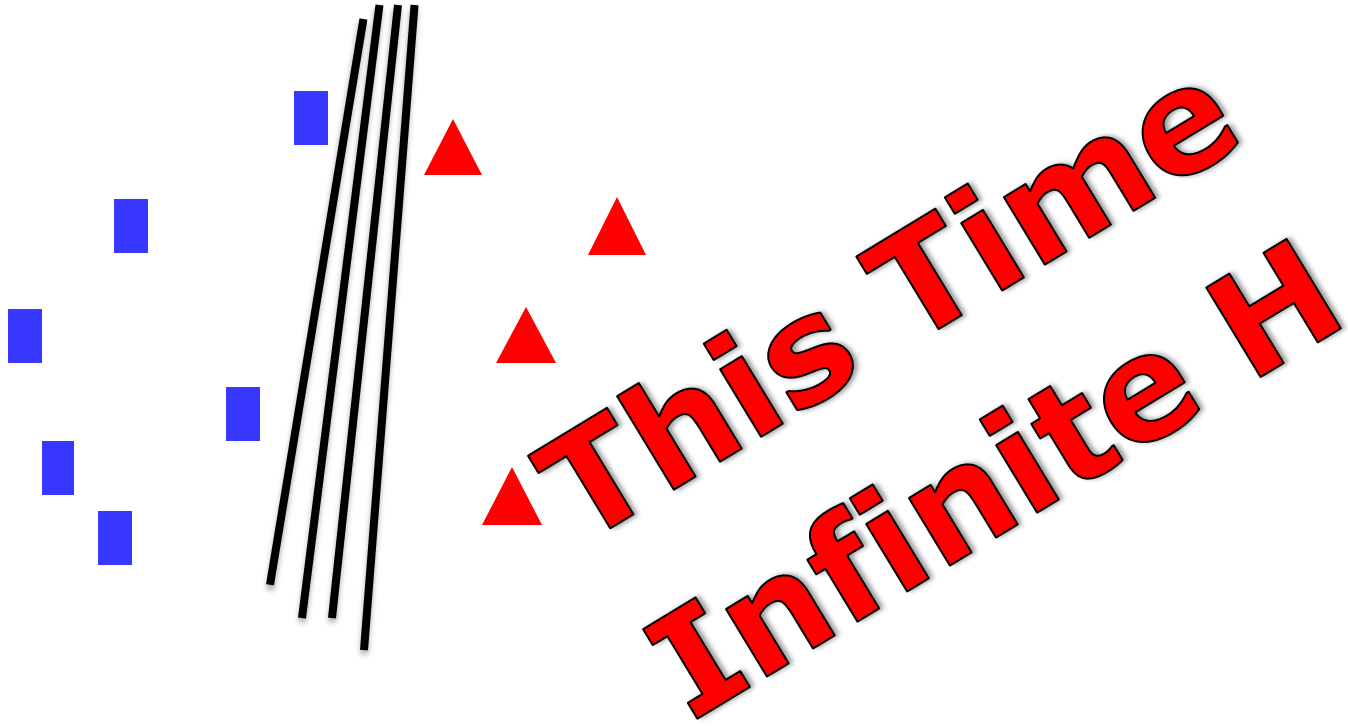
- We proved:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2Me^{-2\varepsilon^2 n}$

- M is |H|, n is #samples.
- h* is hypothesis learned by learning algorithm.

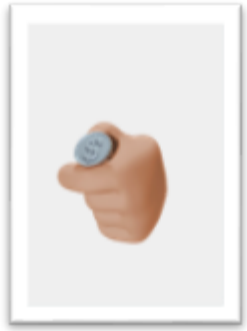
Infinite H



- We proved:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2Me^{-2\varepsilon^2 n}$
- Cannot handle infinite H (M=∞).



Any Hope?



- We proved:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2Me^{-2\varepsilon^2 n}$

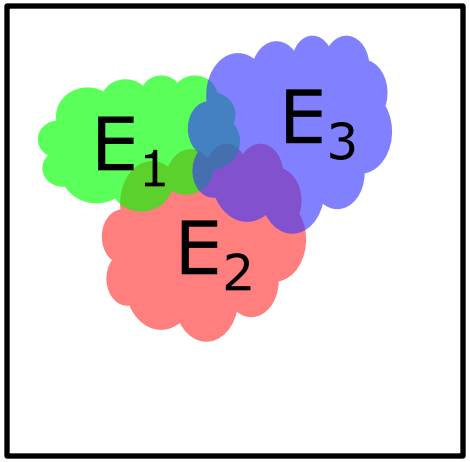
- Do we stand a chance with infinite H?
  - Where M came from:

$\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq$

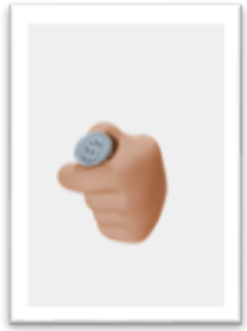
$\Pr[|E_{in}(h_1) - E_{out}(h_1)| > \varepsilon$ **or ... or**
$|E_{in}(h_M) - E_{out}(h_M)| > \varepsilon] \leq$

$\Pr[|E_{in}(h_1) - E_{out}(h_1)| > \varepsilon] + \ldots +$
$\Pr[|E_{in}(h_M) - E_{out}(h_M)| > \varepsilon]$

Proof by picture



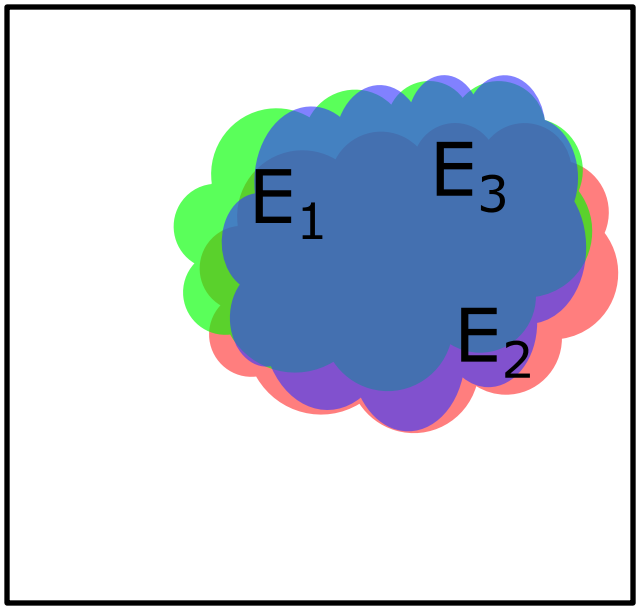
There's Hope!

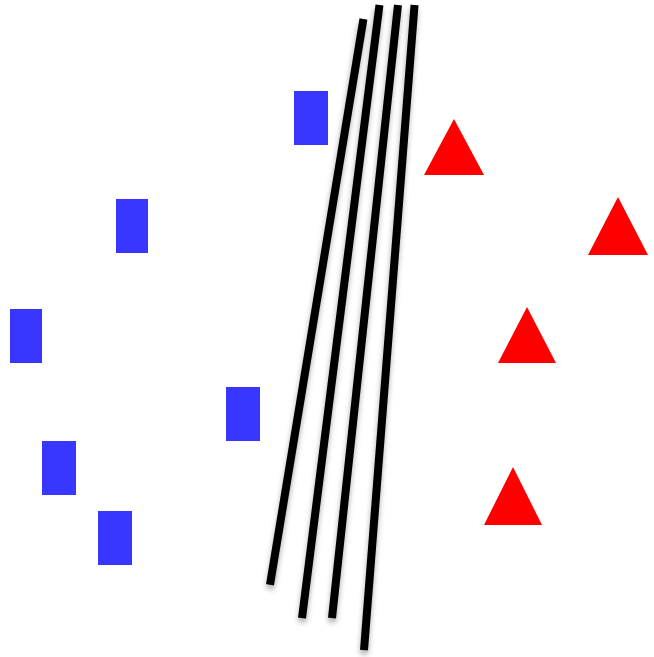


- We proved:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2Me^{-2\varepsilon^2 n}$

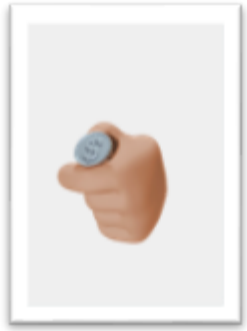
- Union bound very wasteful if hypotheses similar:

  Proof by picture



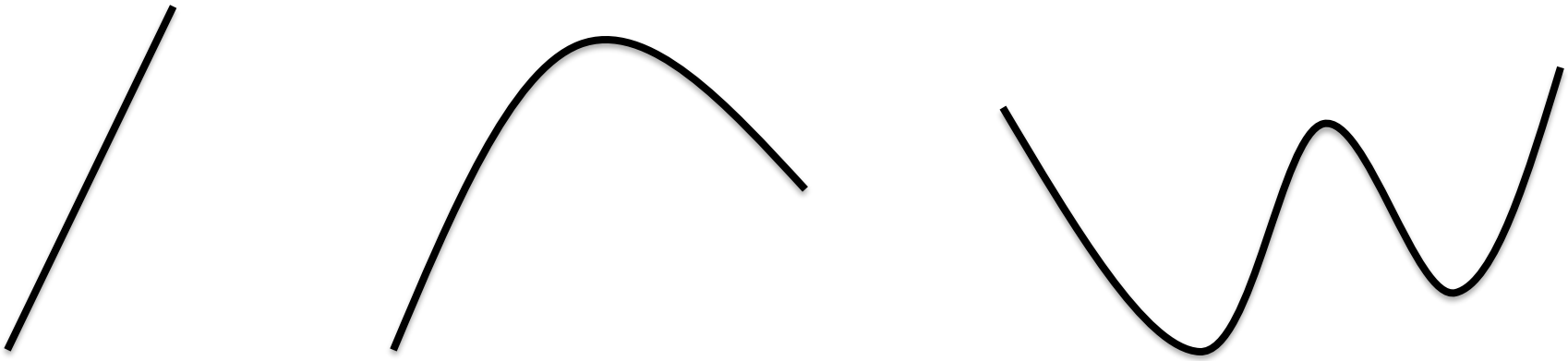


Intuition



- We proved:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2Me^{-2\varepsilon^2 n}$

- Intuition:
  - Exploit similarity among hypotheses.
  - More complex H gives worse generalization.
  - Find different measure than |H| to capture complexity.



The Growth Function

- Complexity of (infinite) H:
  - Growth function $m_H(n)$:

Each distinct labeling called a **dichotomy**

$$m_H(n) = max_{x_1,\dots,x_n \in X} |\{(h(x_1), \dots, h(x_n)) : h \in H\}|$$

Maximum over all sets of n (sample) points in the input domain.

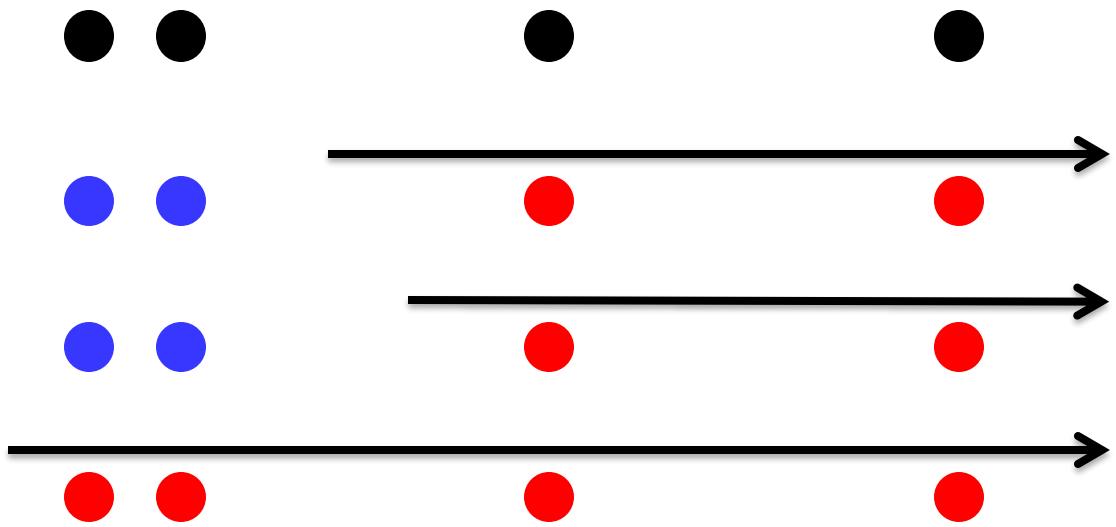
Number of **distinct ways** H can classify the n points.

The Growth Function

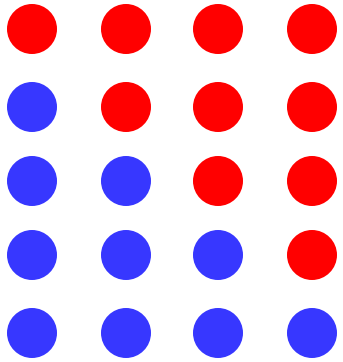
- Growth function

$$m_H(n) = max_{x_1,\dots,x_n \in X} |\{(h(x_1),\dots,h(x_n)) : h \in H\}|$$

- Example X=$\mathbb{R}$, H: all positive rays, n=4:



5 dichotomies



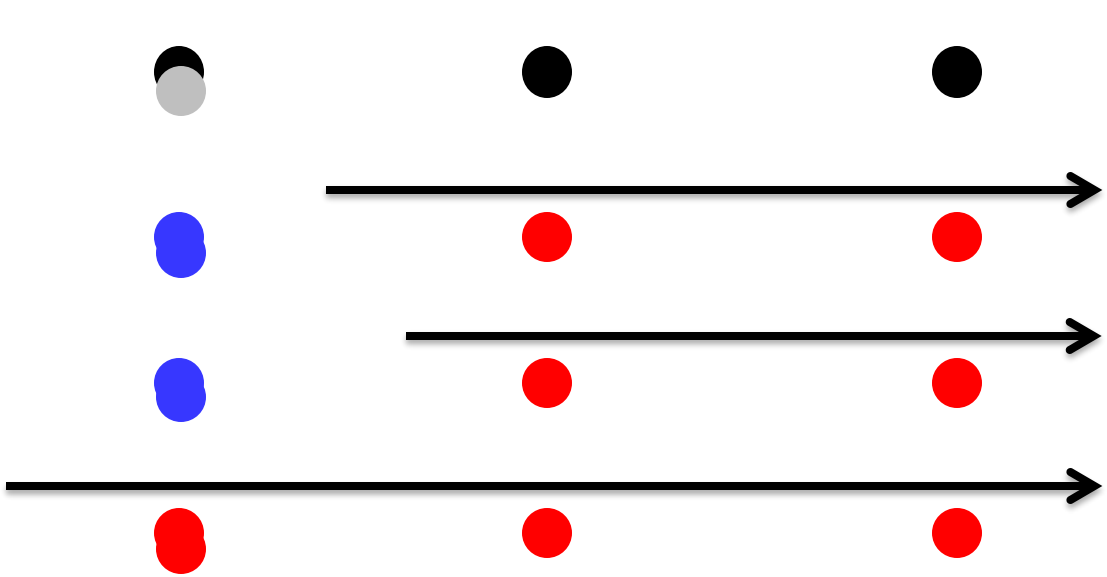
$m_H(4) \geq 5$

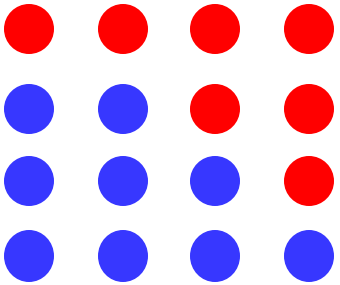
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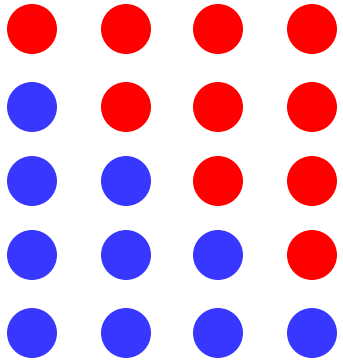
Take max, so still $m_H(4) \geq 5$

The Growth Function

- Growth function

$$m_H(n) = max_{x_1,\dots,x_n \in X} |\{(h(x_1), \dots, h(x_n)) : h \in H\}|$$

- Example X=$\mathbb{R}$, H: all positive rays, n=4:
- $m_H(4) \geq 5$
- $m_H(4) \leq 5$:
  - For any set of 4 points x_1, x_2, x_3, x_4:
  - Order them on the line,
    - break ties arbitrarily.
  - All possible classifications subset of $\Rightarrow$



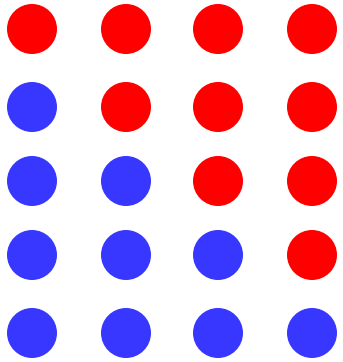
- $m_H(4)=5$

The Growth Function

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- $m_H(n) \leq 1+n$

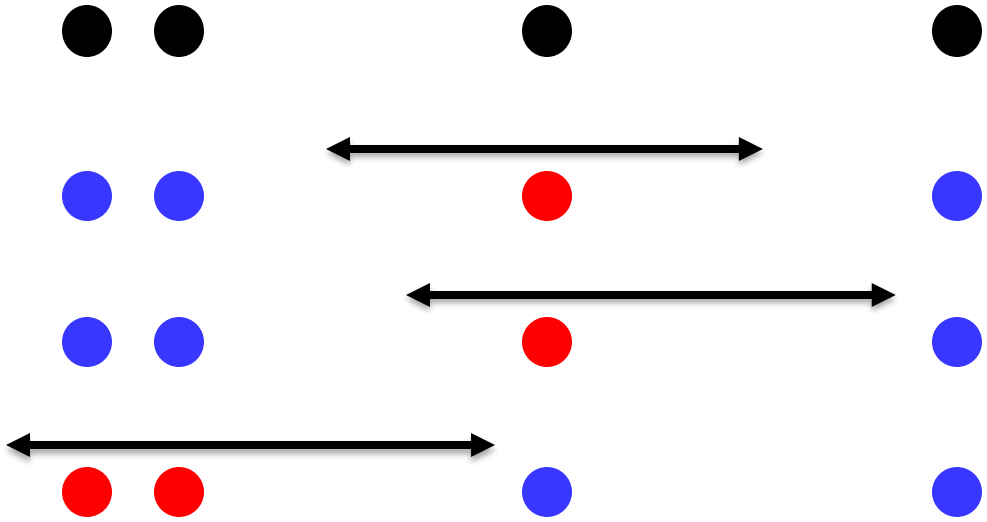


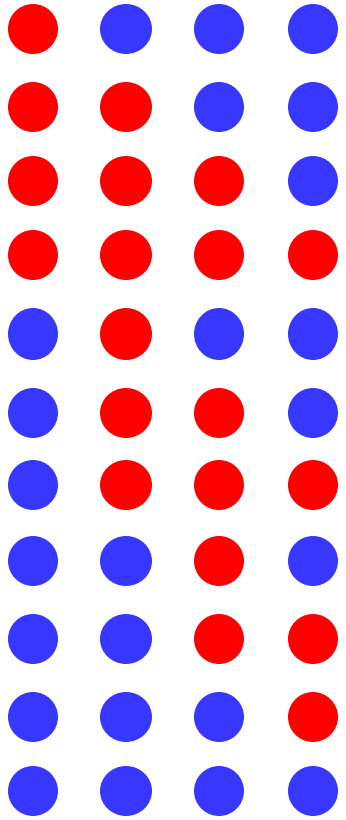
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- Example X=$\mathbb{R}$, H: all intervals, n=4:





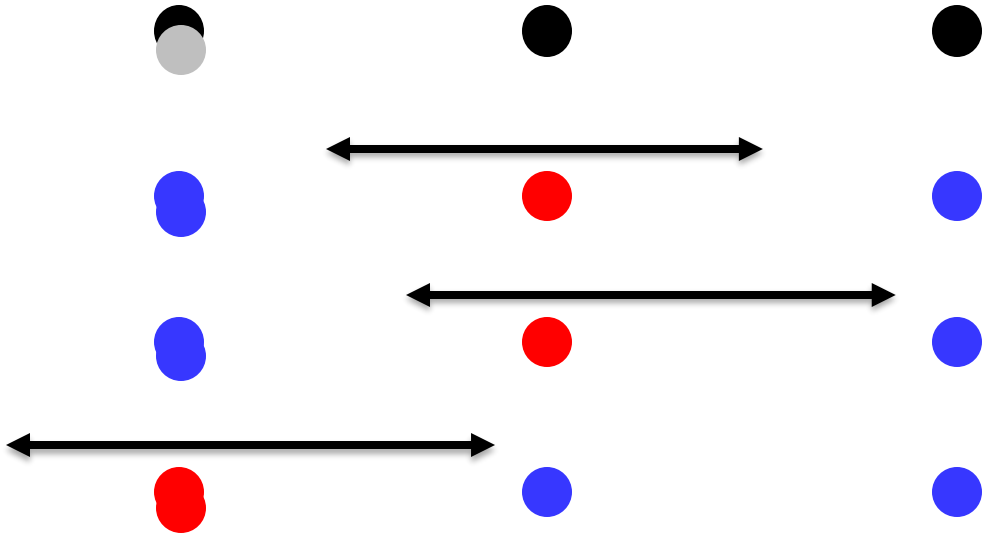
$m_H(4) \geq 11$

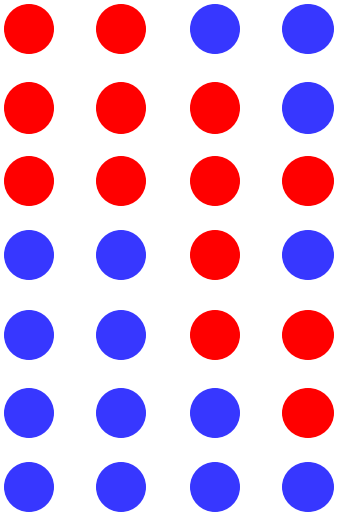
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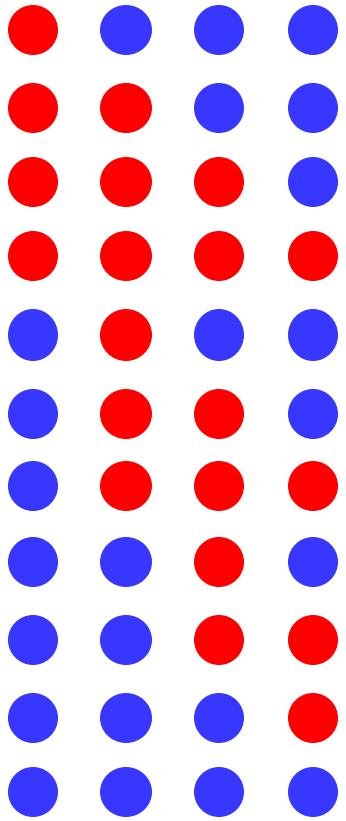
Take max, so still $m_H(4) \geq 11$

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- Example X=$\mathbb{R}$, H: all intervals, n=4:
- $m_H(4) \geq 11$
- $m_H(4) \leq 11$:
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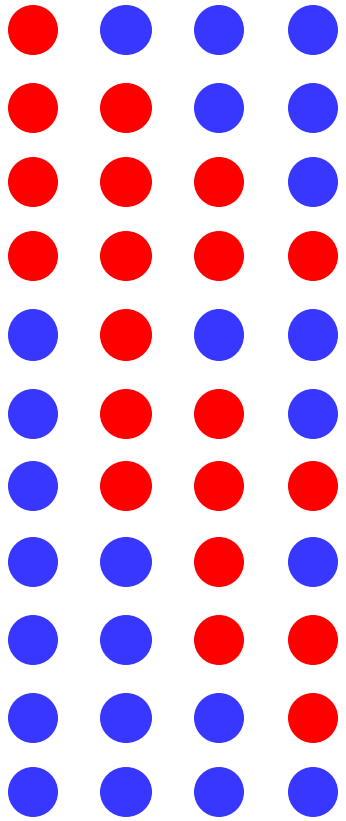


- $m_H(4)=11$

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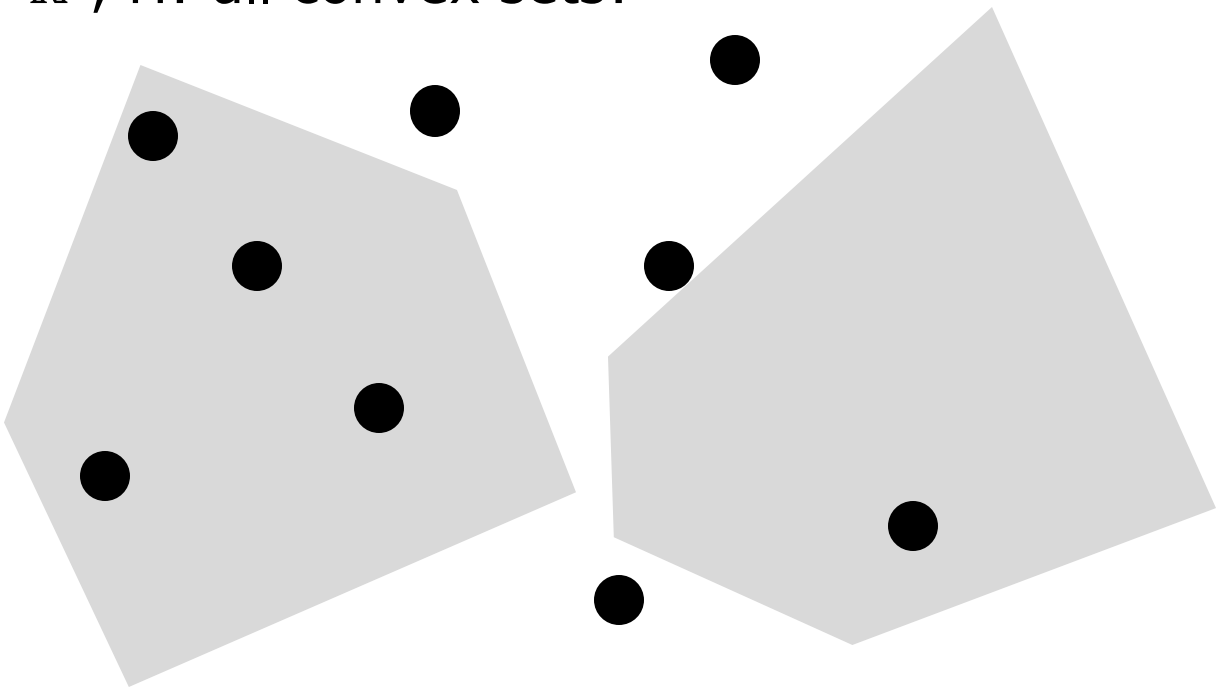
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    - (First included x_i, last included x_j)
    - or empty set.
- $m_H(n) \leq 1+n^2/2+n/2$

The Growth Function

- Growth function

$$m_H(n) = max_{x_1,\dots,x_n \in X} |\{(h(x_1), \dots, h(x_n)) : h \in H\}|$$

- Example X=$\mathbb{R}^2$, H: all convex sets:

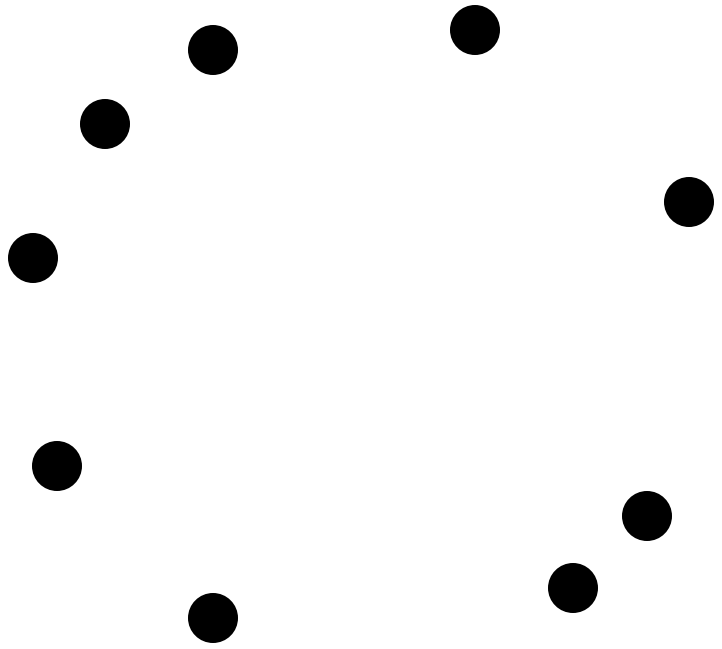


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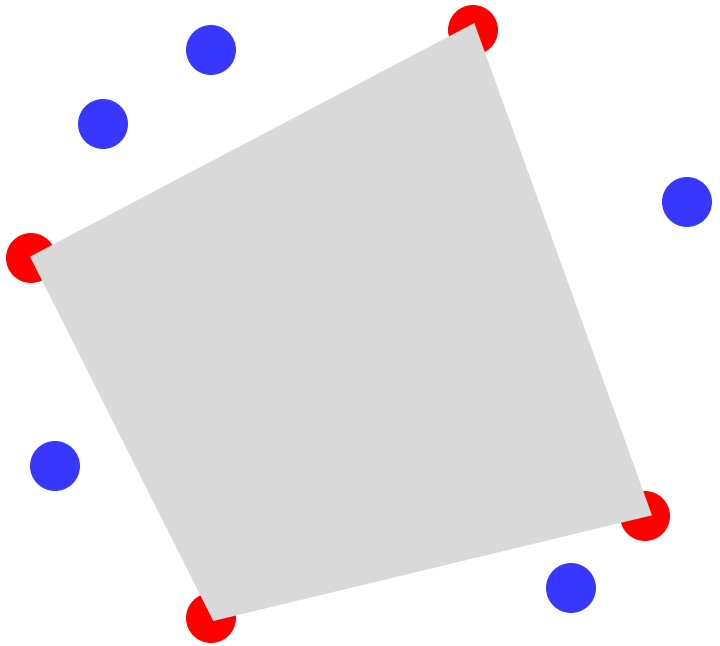


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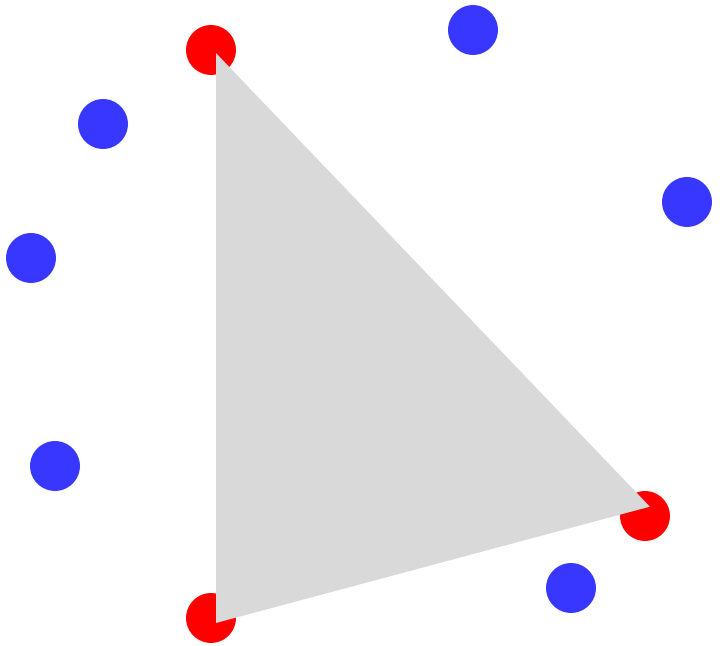


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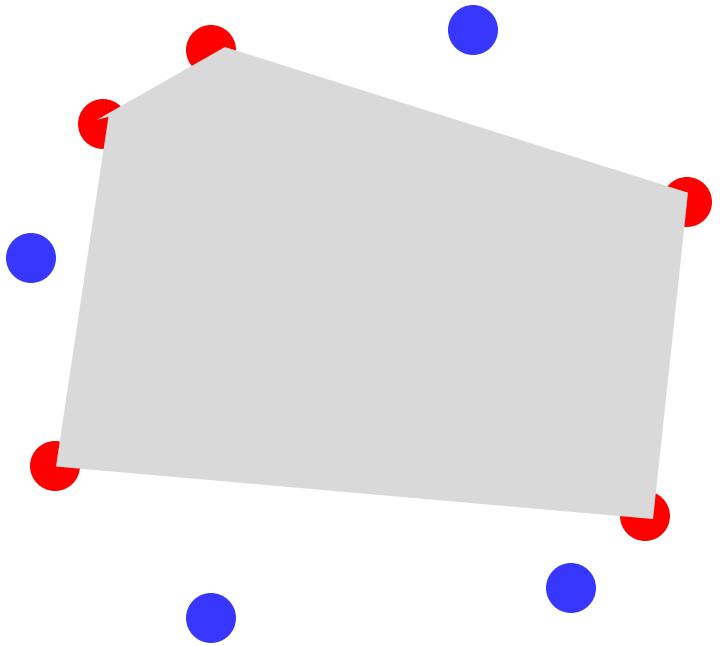


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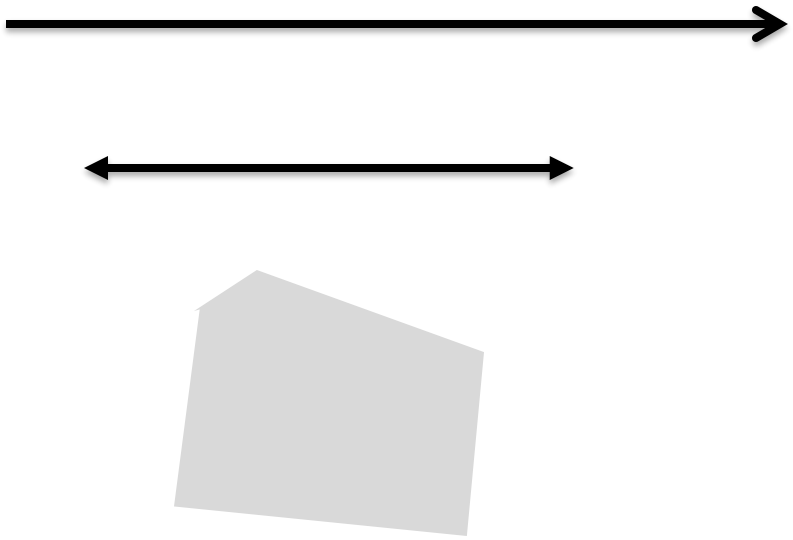
$m_H(n)=2^n$

The Growth Function

- Growth function

$$m_H(n) = max_{x_1,\dots,x_n \in X} |\{(h(x_1), \dots, h(x_n)) : h \in H\}|$$

- Different complexities:
  - Positive rays (1d)
    - $m_H(n) = 1+n$
  - Intervals (1d)
    - $m_H(n) = 1+n^2/2+n/2$
  - Convex sets (2d)
    - $m_H(n) = 2^n$



Generalization via Growth Function

- Growth function

$$m_H(n) = max_{x_1,\dots,x_n \in X} |\{(h(x_1),\dots,h(x_n)) : h \in H\}|$$

- Can we replace our previous bound:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2Me^{-2\varepsilon^2 n}$
- With:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2m_H(n)e^{-2\varepsilon^2 n}$



- At least get something finite!
- Almost true!
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8m_H(2n)e^{-\varepsilon^2 n/8}$

We will **not** give proof

Generalization via Growth Function

- Generalization:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8m_H(2n)e^{-\varepsilon^2 n/8}$
- Where:
  - $m_H(n) = max_{x_1,\dots,x_n \in X}|\{(h(x_1),\dots,h(x_n)) : h \in H\}|$

- Our Examples:
  - Positive rays (1d)
    - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8(1+2n)e^{-\varepsilon^2 n/8}$
  - Intervals (1d)
    - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8(1+2n^2+n)e^{-\varepsilon^2 n/8}$
  - Convex sets (2d)
    - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2^{2n+3}e^{-\varepsilon^2 n/8}$



>1 for all n≥1 and ε≤1

Bounding the Growth Function

- Generalization:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8m_H(2n)e^{-\varepsilon^2 n/8}$
- Where:
  - $m_H(n) = max_{x_1,\dots,x_n \in X}|\{(h(x_1),\dots,h(x_n)) : h \in H\}|$
- Issue:
  - Tedious/unrealistic to find exact formula for $m_H(n)$.
- Observation:
  - We only need upper bound on $m_H(n)$ to use formula!

Learning Theory

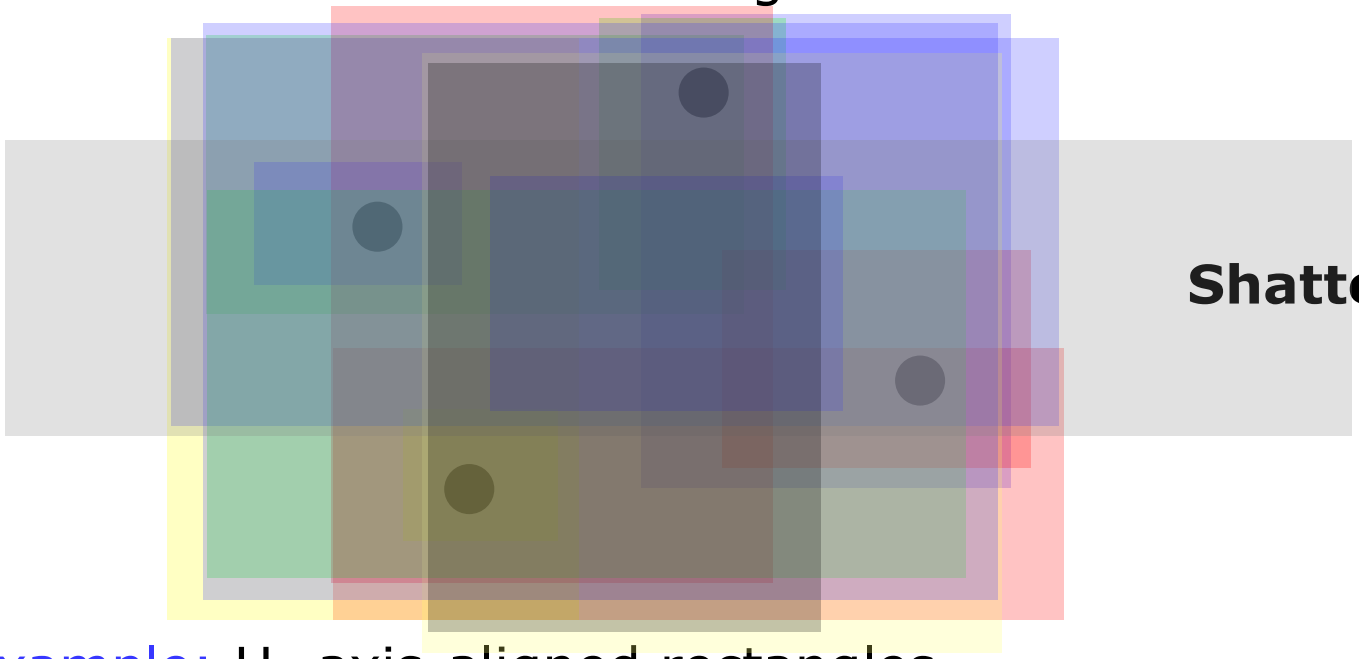
Infinite Hypothesis Sets

Bounding the Growth Function

- Generalization:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8m_H(2n)e^{-\varepsilon^2 n/8}$
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Shattering

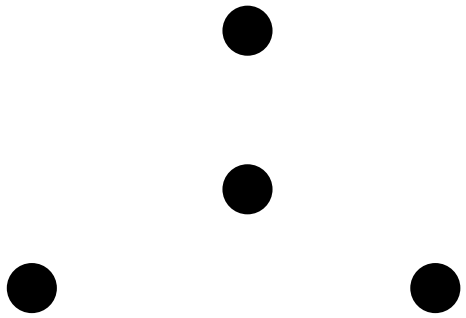
- A step towards bounding $m_H(n)$:
  - A set of k samples $x_1,\ldots,x_k \in X$ is shattered by H if:
    - All 2^k dichotomies can be generated.



  - Example: H=axis-aligned rectangles.

Shattering

- A step towards bounding $m_H(n)$:
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Not shattered

  - Example: H=axis-aligned rectangles.

Break Points

- A step towards bounding $m_H(n)$:
  - If **no** data set of k samples $x_1,\ldots,x_k \in X$ can be shattered by H, then k is a break point for H.



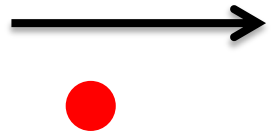
  - k=4 is **NOT** a break point. Proved by left example.
  - **Exercise:** Prove that k=5 is a break point.

Break Points

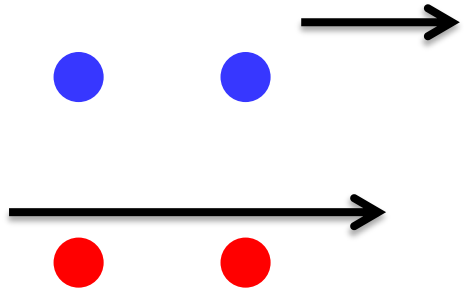
- A step towards bounding $m_H(n)$:
  - If **no** data set of k samples $x_1,\ldots,x_k \in X$ can be shattered by H, then k is a break point for H.

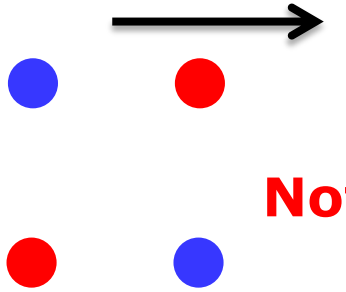
- Example: Positive rays.
  - 1 point?





  - 2 points?





Not possible!

k=2 break point

Break Points

- A step towards bounding $m_H(n)$:
  - If **no** data set of k samples $x_1,\ldots,x_k \in X$ can be shattered by H, then k is a break point for H.

- Example: Intervals.
  - 2 points?

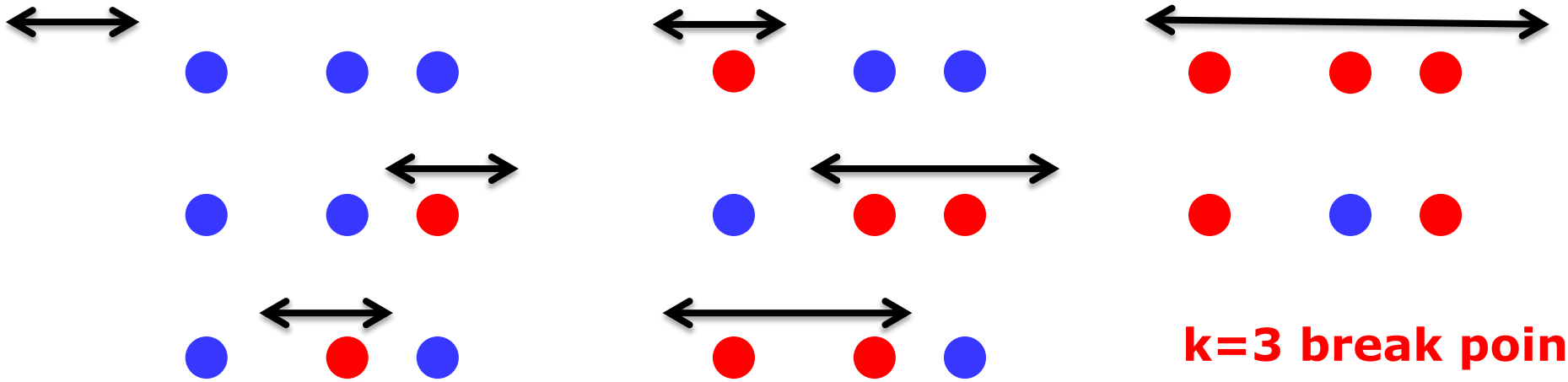


  - k=2 not break point.

Break Points

- A step towards bounding $m_H(n)$:
  - If **no** data set of k samples $x_1,\ldots,x_k \in X$ can be shattered by H, then k is a break point for H.

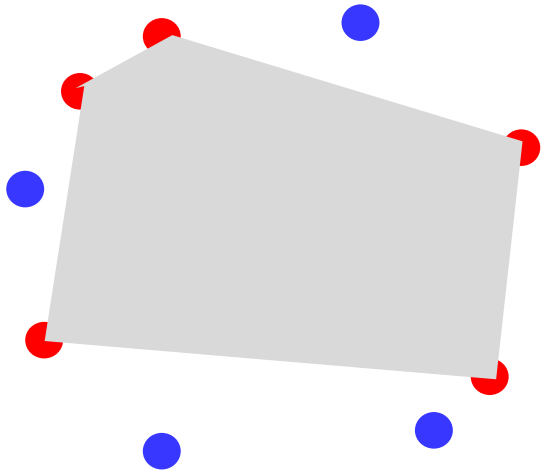
- Example: Intervals.
  - 3 points?



Break Points

- A step towards bounding $m_H(n)$:
  - If **no** data set of k samples $x_1,\ldots,x_k \in X$ can be shattered by H, then k is a break point for H.

- Example: Convex sets in $\mathbb{R}^2$.



- No break points!

VC Dimension

- A step towards bounding $m_H(n)$:
  - If **no** data set of k samples $x_1,\ldots,x_k \in X$ can be shattered by H, then k is a break point for H.

- Vapnik-Chervonenkis dimension (VC dimension):
  - VC-dimension of H is d such that:
    - If k is the smallest break point for H, then d=k-1.

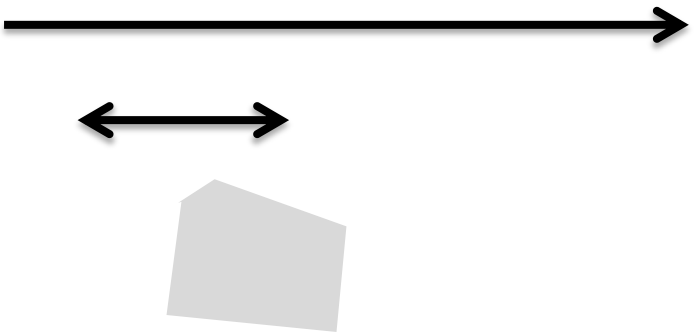
  - Equivalently: d is the largest integer such that there is a data set of size d that can be shattered by H.

VC Dimension

- A step towards bounding $m_H(n)$:
  - If **no** data set of k samples $x_1,\ldots,x_k \in X$ can be shattered by H, then k is a break point for H.

- Vapnik-Chervonenkis dimension (VC dimension):
  - VC-dimension of H is d such that:
    - If k is the smallest break point for H, then d=k-1.

- Examples:
  - Positive rays: VC dimension 1
  - Intervals: VC dimension 2
  - Convex sets: VC dimension ∞



VC Dimension

- Vapnik-Chervonenkis dimension (VC dimension):
  - VC-dimension of H is d such that:
    - If k is the smallest break point for H, then d=k-1.

- Bounds on growth function:
  - Assume H has VC dimension d, then:
    - $m_H(n) \leq n^d+1$

- Generalization:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8m_H(2n)e^{-\varepsilon^2 n/8}$
- Becomes:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8((2n)^d+1)e^{-\varepsilon^2 n/8}$

VC Dimension

- H with VC dimension d has generalization:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8((2n)^d+1)e^{-\varepsilon^2 n/8}$
- Examples:
  - Positive rays
    - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8(2n+1)e^{-\varepsilon^2 n/8}$
  - Intervals
    - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8((2n)^2+1)e^{-\varepsilon^2 n/8}$
  - Convex sets
    - Meaningless

VC Dimension

- H with VC dimension d has generalization:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8((2n)^d+1)e^{-\varepsilon^2 n/8}$
- Observation:
  - n^d grows polynomially with n.
  - $e^{-\varepsilon^2 n/8}$ decreases exponentially with n.
- Conclusions:
  - If VC dimension is bounded, then given enough data, E_{in} and E_{out} will be close.

VC Dimension

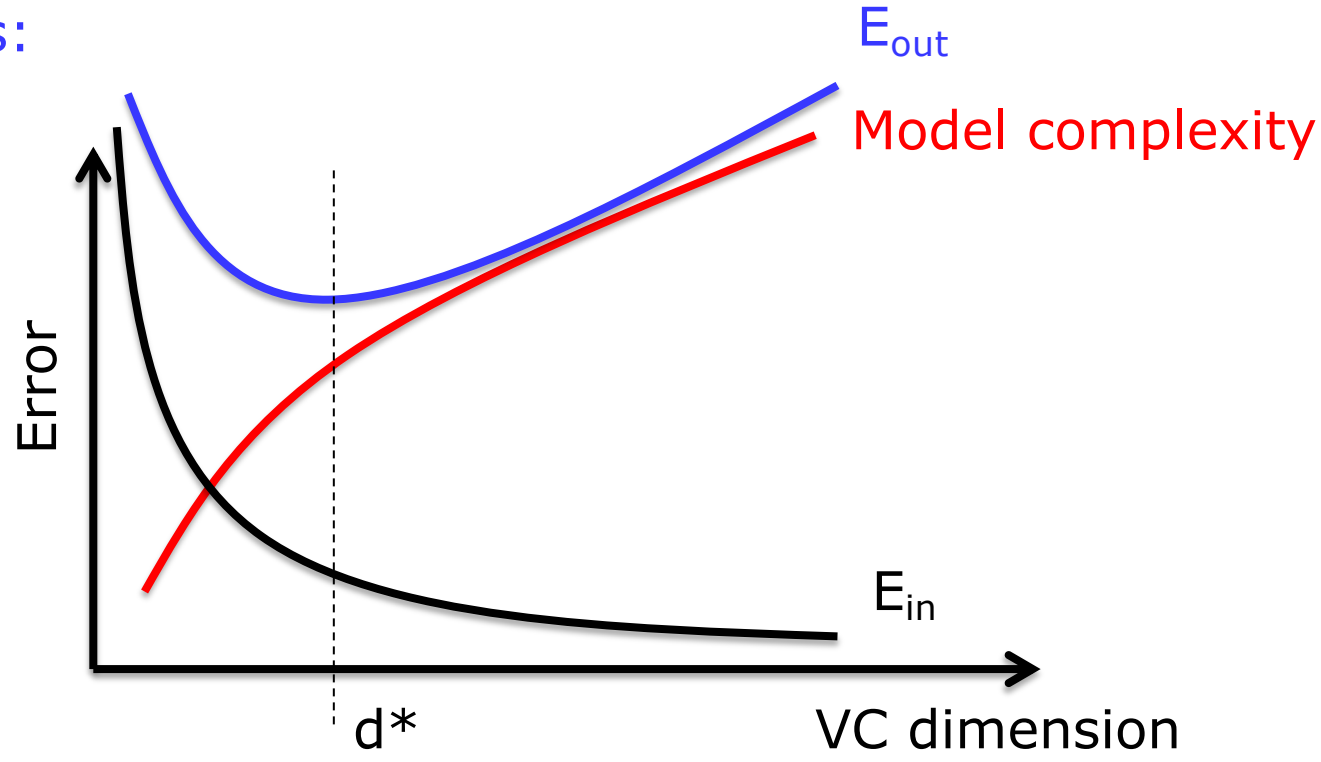
- H with VC dimension d has generalization:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8((2n)^d+1)e^{-\varepsilon^2 n/8}$

- Conclusions:
  - VC dimension easier to bound than getting exact formula for growth function.
  - Gives a concrete way to measure complexity of hypothesis set.
  - With less complex hypothesis set, we can be more certain that E_{in} and E_{out} are close.

VC Dimension

- H with VC dimension d has generalization:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8((2n)^d+1)e^{-\varepsilon^2 n/8}$

- Conclusions:



Learning Theory

Testing

Recap

- Last Time:
  - VC-dimension:
    - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8((2n)^d + 1)e^{-\varepsilon^2 n/8}$

- Insights:
  - More data means E_{in} better predictor of E_{out}.
  - More complex model H means E_{in} worse predictor of E_{out}.
  - Minimizing E_{in} is a good strategy when given enough data.

Using the VC-bound

- Use case:
  - You train a machine learning model for a customer.
  - Customer would like an estimate of E_{out} that is accurate to within 0.1 with 90% confidence.
  - How much data do you need?
    - You know E_{in} and that the hypothesis set you used has VC-dimension 25.

- Use:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8((2n)^d+1)e^{-\varepsilon^2 n/8}$

Using the VC-bound

- Have d=25, ε=0.1, want 90% confidence.
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8((2n)^d+1)e^{-\varepsilon^2 n/8}$
- Enough data if:
  - $8((2n)^d+1)e^{-\varepsilon^2 n/8} \leq 0.1 \leftarrow$
  - $8((2n)^{25}+1)e^{-n/800} \leq 0.1$
- Try some guesses for n. n=8'000:
  - $8\left((16'000)^{25}+1\right)e^{-10} \leq 0.1 \leftarrow 8(e^{242}+1)e^{-10} \leq 0.1$
- n=80'000:
  - $8\left((160'000)^{25}+1\right)e^{-100} \leq 0.1 \leftarrow 8(e^{300}+1)e^{-100} \leq 0.1$
- n=400'000
  - $8\left((800'000)^{25}+1\right)e^{-500} \leq 0.1 \leftarrow 8(e^{340}+1)e^{-500} \leq 0.1$

Using the VC-bound

- Use case:
  - You train a machine learning model for a customer.
  - Customer would like an estimate of E_{out} that is accurate to within 0.1 with 90% confidence.
  - You know E_{in} and that the hypothesis set you used has VC-dimension 25.

- VC-bound:
  - Need somewhere between n=80'000 and n=400'000!
  - Very pessimistic!
- VC-bound far off in practice.
  - Around a factor 1'000 in terms of data items n.

Using the VC-bound

- Reason VC-bound is off?
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8((2n)^d+1)e^{-\varepsilon^2 n/8}$
- Wrong asymptotic formula?
  - No, there exists examples where tight.
- Real data often easier than worst case.
- Constant factors.

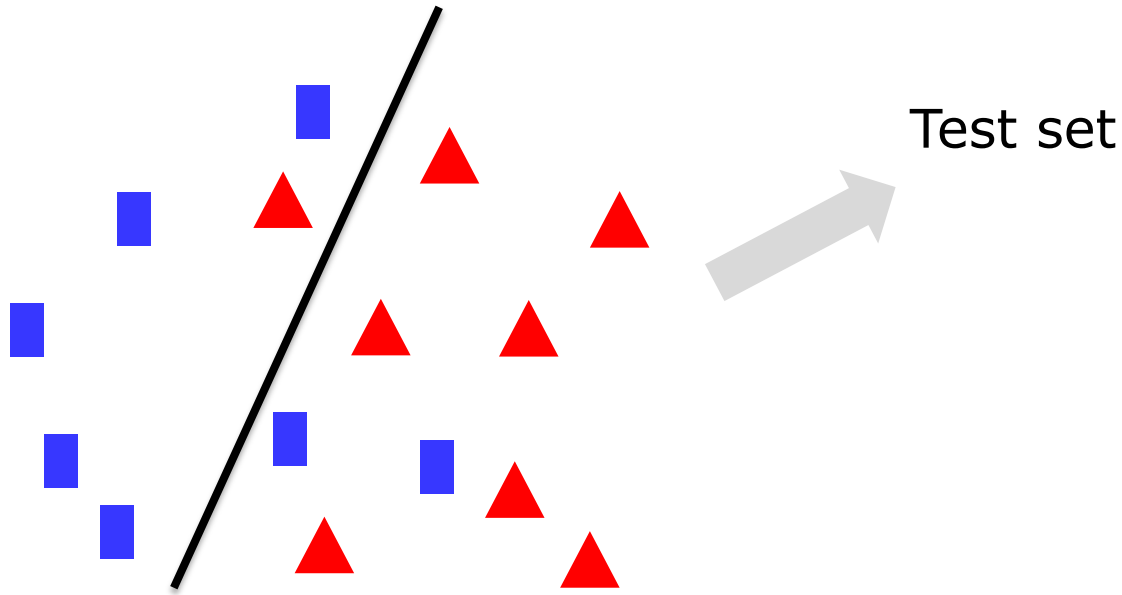
- Rule of thumb: If want $E_{out} \leq E_{in}+0.1$ with 90% confidence, then need about n=10d data.

- What good is the VC-bound?
  - We gained intuition on the effect of more complex hypothesis sets.

Testing

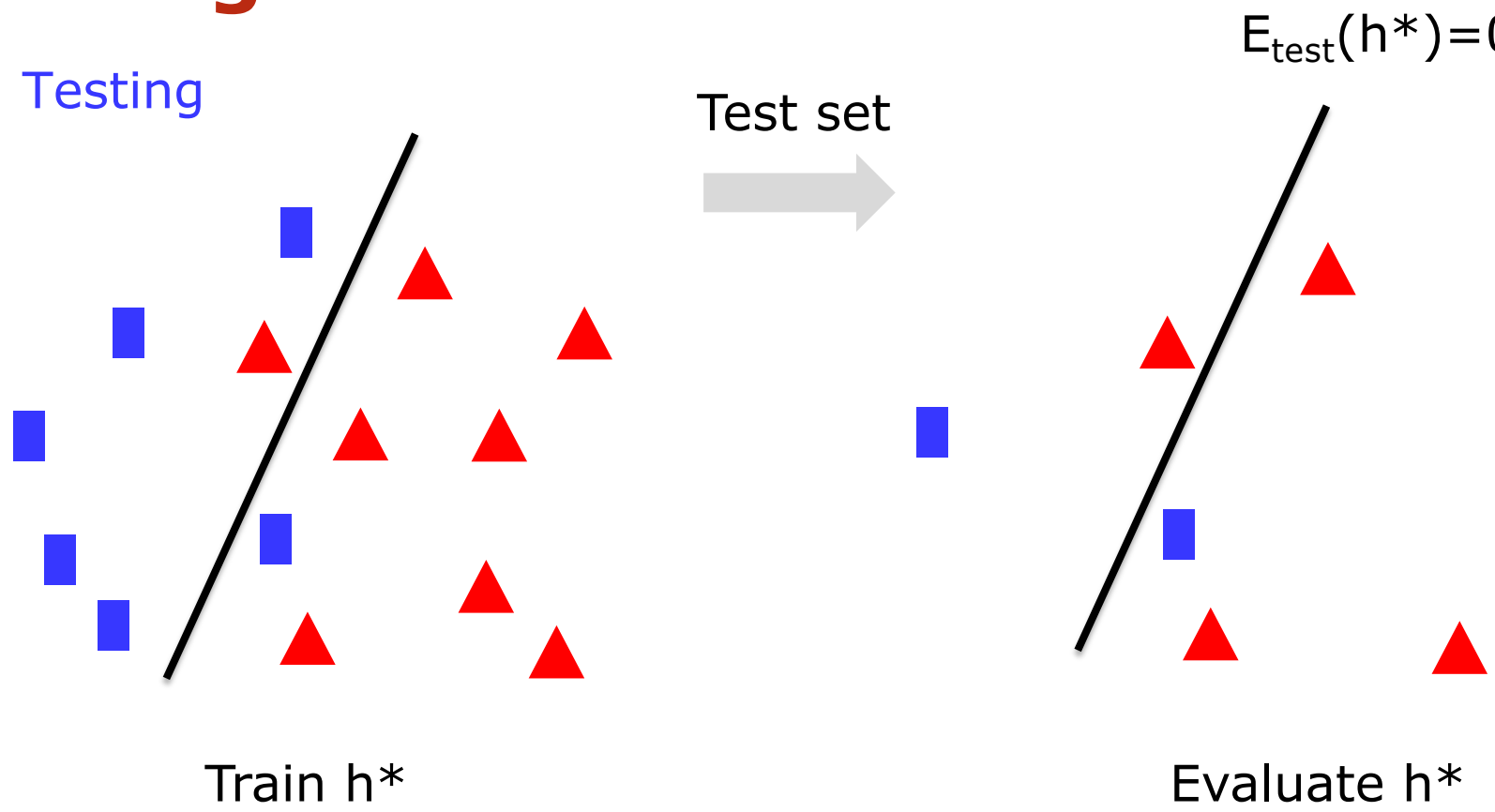
- Use case:
  - You train a machine learning model for a customer.
  - Customer would like an accurate estimate of E_{out} with good confidence.
- What else can we do?

- Take some of data and make it a test set
- Train on remaining
- Estimate E_{out} from test set.



Testing

- Testing

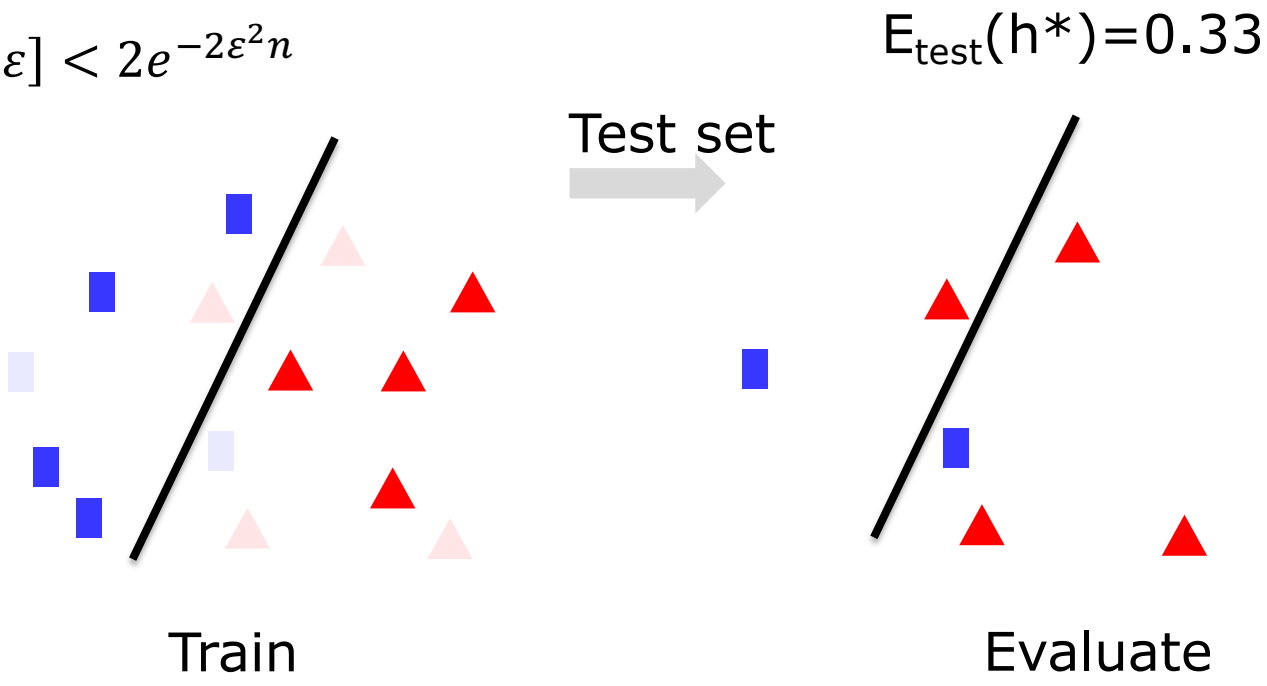


Testing

- How accurate is $E_{test}(h^*)$?
- Can we give any guarantees?

- Hoeffding bound?
- $\Pr[|E_{in}(h) - E_{out}(h)| > \varepsilon] < 2e^{-2\varepsilon^2 n}$

- Assumptions:
- h chosen independently of data.

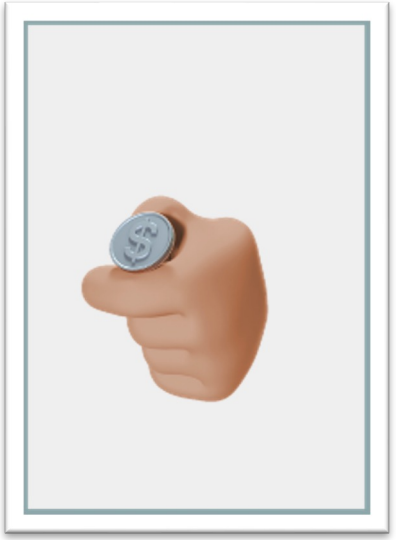


Testing

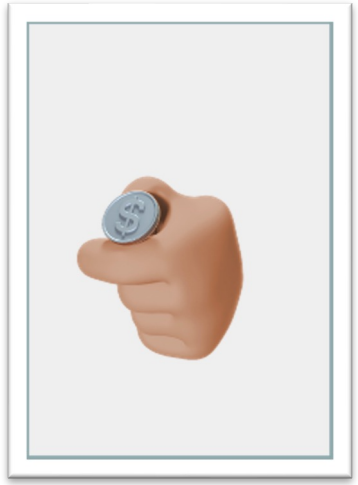
- How accurate is $E_{test}(h^*)$?
- Can we give any guarantees?

- Recall:
  - Assume samples generated i.i.d. from distribution D.
  - For a fixed hypothesis h:
    - Flip n coins with Pr[heads]=$E_{out}(h)$.
    - $E_{in}(h)$=fraction of heads.

  - Hoeffding then gave
    - $\Pr[|E_{in}(h) - E_{out}(h)| > \varepsilon] < 2e^{-2\varepsilon^2 n}$



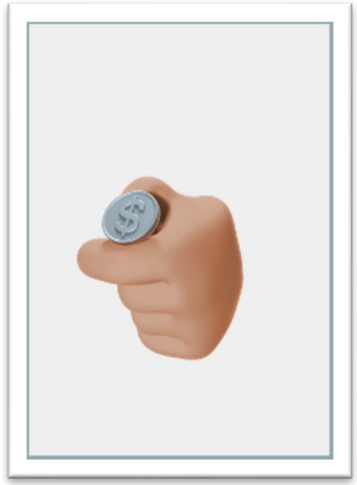
Testing



- How accurate is $E_{test}(h^*)$?
- Can we give any guarantees?

- Recall:
  - Assume samples generated i.i.d. from distribution D.
- For Testing:
  - If we choose the test set without looking at the data, e.g. a random subset of k items or the first k items, then:
    - Remaining n-k still i.i.d. samples from D.
    - The k test items are also i.i.d. samples from D.
- Implications:
  - If h* is trained without test set, then Hoeffding still applies!

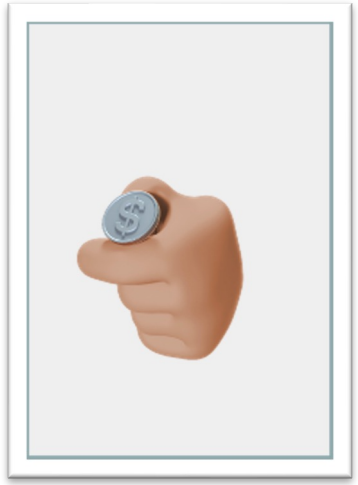
Testing



- How accurate is $E_{test}(h^*)$?
- Can we give any guarantees?

- Test set of size k.
  - For a hypothesis h* chosen without looking at test set:
    - Flip k coins with Pr[heads]=$E_{out}(h^*)$.
    - $E_{test}(h^*)$=fraction of heads.
  - Hoeffding gives
    - $\Pr[|E_{test}(h^*) - E_{out}(h^*)| > \varepsilon] < 2e^{-2\varepsilon^2 k}$
- Intution: the test set might as well have been sampled after choosing the hypothesis h*.

Testing



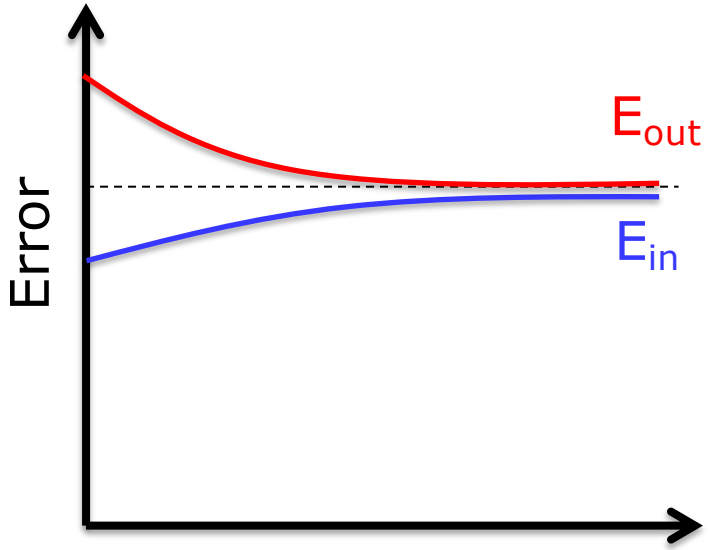
- Comparison:
  - $\Pr[|E_{test}(h^*) - E_{out}(h^*)| > \varepsilon] < 2e^{-2\varepsilon^2 k}$
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] < 2Me^{-2\varepsilon^2 n}$
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] < 8((2n)^d+1)e^{-\varepsilon^2 n/8}$

- Testing is much more accurate.
- You test the actual hypothesis you return.
- Other bounds give a pessimistic guarantee that holds for all possible hypotheses (recall the union bound).

- Downside: Less data for training. Balance between k and n.

Learning Curve

- VC-bound:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] < 8((2n)^d+1)e^{-\varepsilon^2 n/8}$

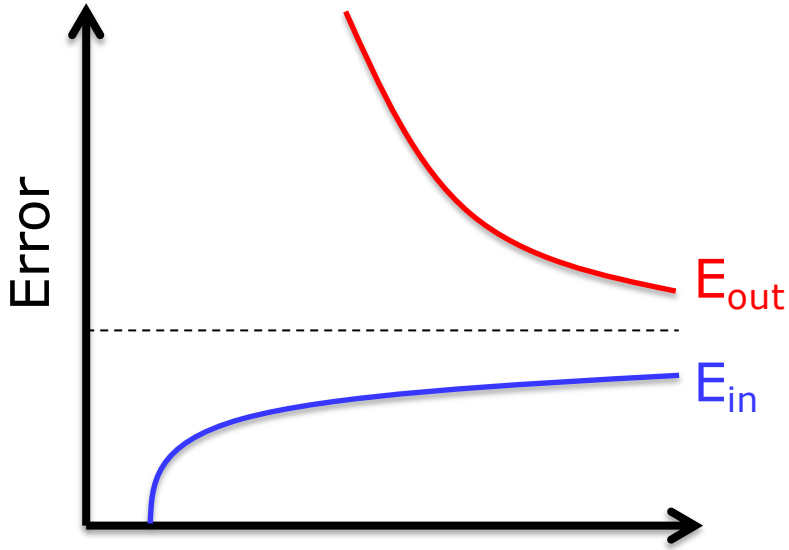


#Samples:

Simple Model

Small E_{out}-E_{in}

Large E_{in}



Complex Model

Large E_{out}-E_{in}

Small E_{in}